MATHS MANTHAN
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Subject: Mathematics | **Weekly Test Paper 08** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **10 questions** divided into **Three Sections**: A, B, and C.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **4 questions** (Q6 of **2 marks**, Q7–Q9 of **3 marks** each).
- **Section C** comprises **1 Applied Case Study** of **4 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. The sum of the first n positive natural numbers is given by the formula: [1]

(A) $\frac{n(n-1)}{2}$ (B) $\frac{n(n+1)}{2}$ (C) $n(n+1)$ (D) $\frac{n^2}{2}$

Q2. The sum of the first n odd positive natural numbers is: [1]

(A) $2n-1$ (B) n^2 (C) $2n^2$ (D) $n(n+1)$

Q3. The sum of the first n positive even integers is: [1]

(A) n^2 (B) $n(n+1)$ (C) $n(n-1)$ (D) $\frac{n(n+1)}{2}$

Q4. If the 9th term of an AP is zero, then the ratio of its 29th term to its 19th term is: [1]

(A) 1 : 2 (B) 2 : 1 (C) 3 : 1 (D) 1 : 3

Q5. **Assertion (A):** If the nth term of a sequence is $a_n = 2n^2 + 1$, then the sequence is an AP.
Reason (R): A sequence is an arithmetic progression if and only if its nth term is a linear expression in n. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Short Answer Questions (VSA & SA) (1 × 2 + 3 × 3 = 11 Marks)

Q6. Find the sum of all natural numbers between 100 and 500 which are exactly divisible by 8. [2]

Q7. If S_1, S_2, S_3 denote the sums of $n, 2n, 3n$ terms respectively of an arithmetic progression, prove that:

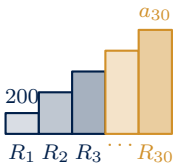
$$S_3 = 3(S_2 - S_1)$$

[3]

Q8. Find four numbers in an arithmetic progression whose sum is 20 and the sum of whose squares is 120. [3]

Q9. The first term of an AP is 5, the last term is 45, and the sum of all its terms is 400. Find the number of terms n and the common difference d. [3]

Q10. CASE STUDY: Tiered Grandstand Seating in a Sports Stadium



A newly built multi-purpose indoor sports stadium features a modern tiered grandstand designed to offer spectators unobstructed sightlines of the playing court.
The first row directly adjoining the court contains 200 spectator seats. To accommodate increasing arena elevation, each subsequent tier contains 20 additional seats more than the row immediately in front of it. The grandstand consists of 30 concentric seating rows in total.

Based on the above architectural scenario, answer the following questions:

(i) Write down the arithmetic progression representing the number of seats from Row 1 to Row 30, and identify the first term a and common difference d. [1]

(ii) Find the total number of spectator seats installed in the 15th row. [1]

(iii) Calculate the total seating capacity of the entire 30-row stadium grandstand. [2]

OR

(iv) If premium tickets in Rows 1 to 10 are priced at Rs. 500 per seat, calculate the total ticketing revenue generated when all seats in these first 10 rows are sold out. [2]

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